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Semi-solid extrusion 3D printing of lamivudine-zidovudine tablets: formulation design, structural characterization, and tunable release performance determination

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Abstract:	Human immunodeficiency virus (HIV) infection remains a major global health challenge, and flexible manufacturing approaches may support tailored antiretroviral therapy. Lamivudine and zidovudine are widely used nucleoside reverse transcriptase inhibitors, yet their incorporation into Semi-solid extrusion (SSE) 3Dprinted dosage forms has not been previously explored. In this study, combination tablets were fabricated using optimized semisolid inks composed of Carbopol 974P NF, gelatinized corn starch, and Hydroxypropyl methylcellulose acetate succinate (HPMCAS MF). The use of HPMCAS in SSEbased 3D printing has not been previously reported, introducing novelty in semisolid ink development for antiretroviral tablets. The inks were evaluated for viscosity, shearthinning behavior, viscoelastic properties, and thixotropic recovery, confirming suitability for extrusionbased printing. Printed tablets exhibited high shape fidelity, structural integrity after lyophilization, and consistent mechanical performance, including weight, dimensions, friability, and disintegration behavior. ATR FTIR analysis verified successful drug incorporation within the polymeric matrix and the absence of detrimental interactions. Coated and uncoated tablets demonstrated distinct release profiles, with Eudragit L100 55 coatings enabling modulation of drug release during gastrointestinal pH transition.,A concise HPLC-DAD method was developed and validated to support formulation characterization, ensuring selective and reliable quantification of both APIs. The method demonstrated ICHcompliant linearity, accuracy, and precision, and was applied to determine drug loading, extraction efficiency, and dissolution samples. Stability and solubility assessments confirmed suitability for formulationspecific quality control.,Overall, this work demonstrates the feasibility of producing lamivudine–zidovudine tablets via SSE 3D printing and highlights the potential of semisolid extrusion and polymeric coatings for designing customizable antiretroviral dosage forms with tunable release characteristics.

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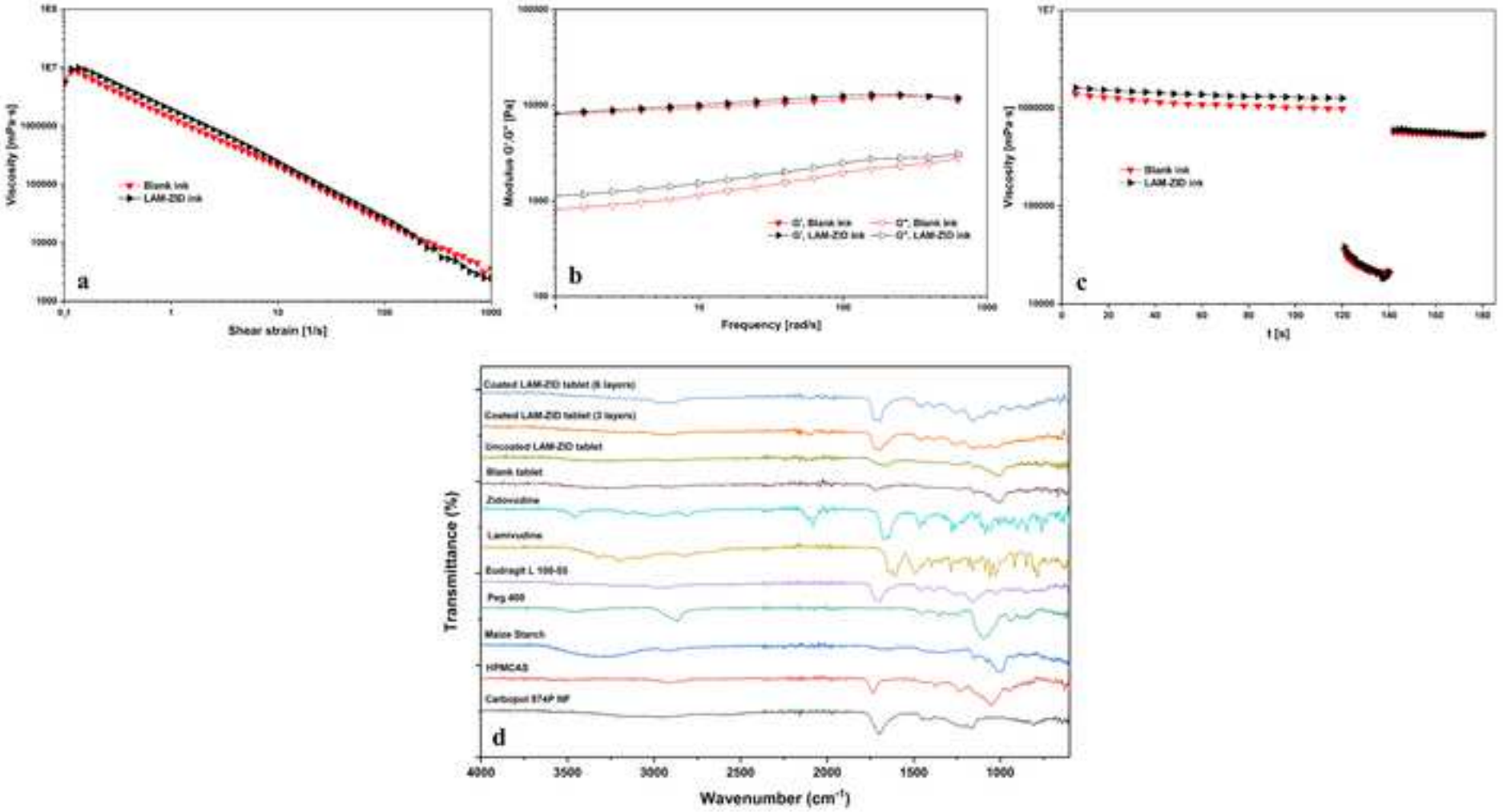
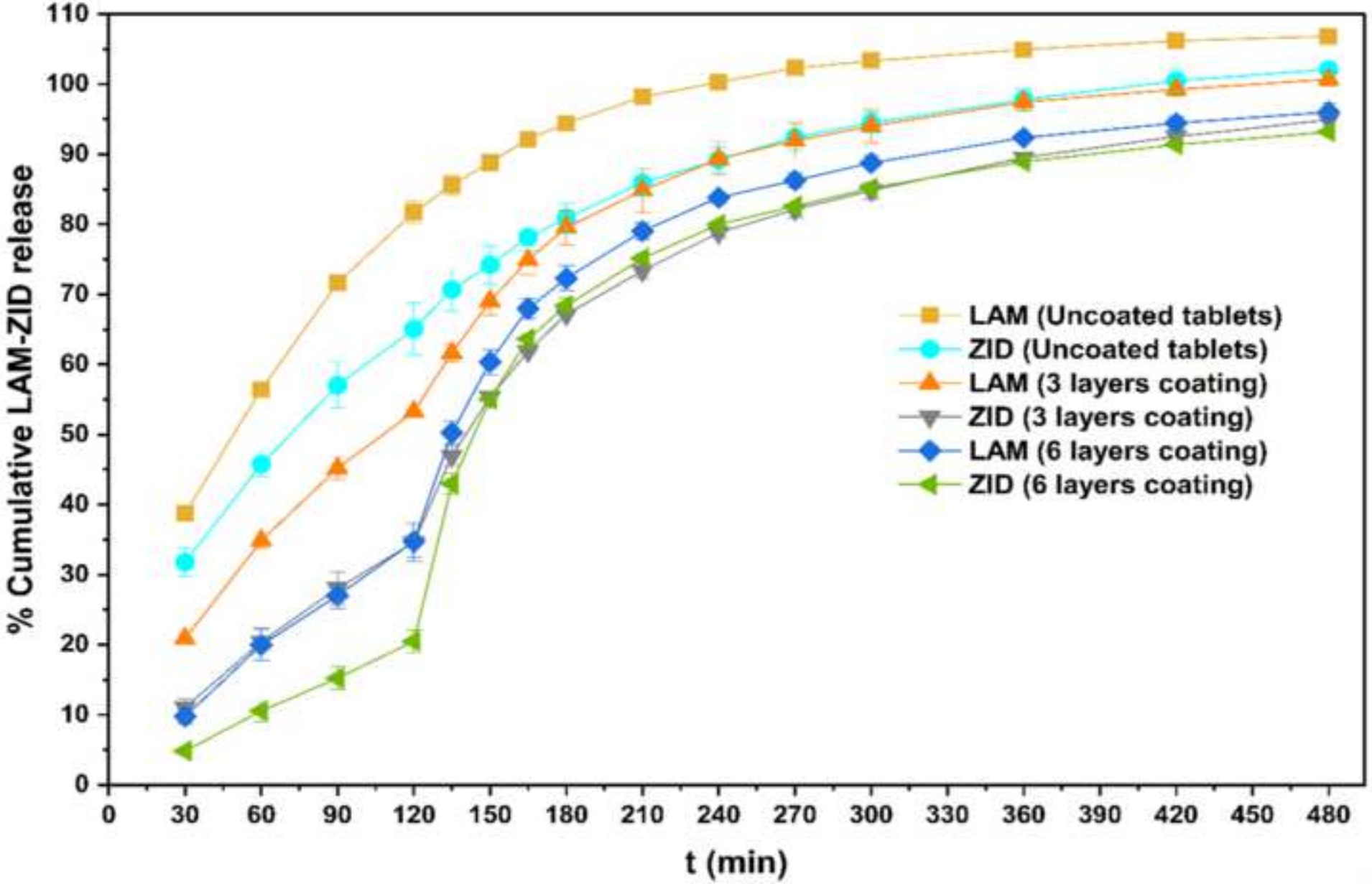


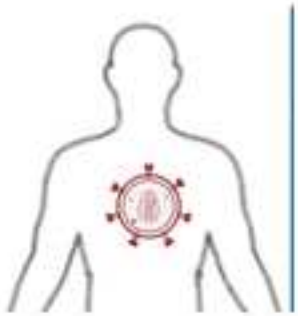
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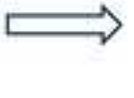
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Semi-solid extrusion (SSE) 3D printing of lamivudine-zidovudine tablets: formulation design, structural characterization, and tunable release performance determination



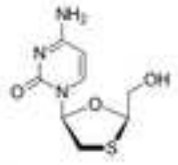
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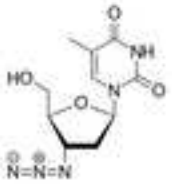








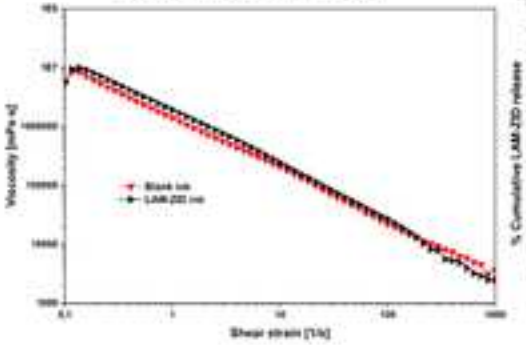
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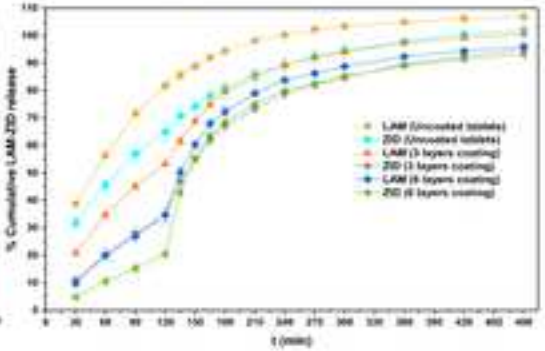
Zidovudine

SSE 3D printing enables customizable lamivudine-zidovudine tablets with robust structure and tunable release

Rheological studies



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Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled **"Semi-solid extrusion 3D printing of lamivudine-zidovudine tablets: formulation design, structural characterization, and tunable release performance determination"** for consideration for publication as a Research Article in the *International Journal of Pharmaceutics*.

In this study, we report the successful development of lamivudine (LAM)–zidovudine (ZID) combination tablets using semi-solid extrusion (SSE) three-dimensional printing, highlighting the potential of this manufacturing approach for the development of customizable antiretroviral dosage forms. Particular emphasis was placed on the optimization of polymer-based semi-solid inks, the characterization of the resulting 3D-printed tablets, and the modulation of drug release through the application of polymeric coatings.

The main novelty of this work lies in the incorporation of hydroxypropyl methylcellulose acetate succinate (HPMCAS) into SSE-based 3D printing formulations, which, to the best of our knowledge, has not been previously reported. Following systematic formulation optimization and rheological characterization, a combination of Carbopol 974P NF, gelatinized corn starch, and HPMCAS was identified as suitable for extrusion-based printing. The resulting tablets exhibited satisfactory printability, shape fidelity, structural integrity after lyophilization, and appropriate physicochemical and mechanical characteristics.

Furthermore, the application of Eudragit L100-55 coatings enabled the modification of the *in vitro* release profiles of both active pharmaceutical ingredients, demonstrating the possibility of adjusting drug release through the number of coating layers. The experimental release data were additionally evaluated using established kinetic models, providing further insight into the release behavior of the developed formulations.

An additional contribution of this work is the development and validation of a reliable HPLC-based analytical method for the simultaneous determination of LAM and ZID. The validated method demonstrated satisfactory selectivity, linearity, accuracy, and precision, supporting the quantitative evaluation of drug content and *in vitro* dissolution performance.

We believe that our findings are particularly relevant to the readership of the *International Journal of Pharmaceutics*, as they combine innovative pharmaceutical formulation development, additive manufacturing, polymer-based drug delivery systems, and analytical quality assessment. The proposed approach provides a foundation for further research into customizable combination dosage forms with adjustable release characteristics.

We confirm that this manuscript represents original work, has not been published previously, and is not under consideration for publication elsewhere. All authors have reviewed and approved the submitted manuscript and agree with its submission to the *International Journal of Pharmaceutics*. Any relevant competing interests and funding information are disclosed in the submission documents.

The supplementary material submitted should be included for review and be published with the

article should the manuscript be accepted for publication.

Thank you for considering our manuscript for publication. We look forward to your response.

Yours faithfully,

Chatzitaki Aikaterini Theodora

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Semi-solid extrusion 3D printing of lamivudine-zidovudine tablets: formulation design, structural characterization, and tunable release performance determination

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Human immunodeficiency virus (HIV) infection remains a major global health challenge, and flexible manufacturing approaches may support tailored antiretroviral therapy. Lamivudine and zidovudine are widely used nucleoside reverse transcriptase inhibitors, yet their incorporation into Semi-solid extrusion (SSE) 3D-printed dosage forms has not been previously explored. In this study, combination tablets were fabricated using optimized semi-solid inks composed of Carbopol 974P NF, gelatinized corn starch, and Hydroxypropyl methylcellulose acetate succinate (HPMCAS MF). The use of HPMCAS in SSE-based 3D printing has not been previously reported, introducing novelty in semi-solid ink development for antiretroviral tablets. The inks were evaluated for viscosity, shear-thinning behavior, viscoelastic properties, and thixotropic recovery, confirming suitability for extrusion-based printing. Printed tablets exhibited high shape fidelity, structural integrity after lyophilization, and consistent mechanical performance, including weight, dimensions, friability, and disintegration behavior. ATR FTIR analysis verified successful drug incorporation within the polymeric matrix and the absence of detrimental interactions. Coated and uncoated tablets demonstrated distinct release profiles, with Eudragit L100 55 coatings enabling modulation of drug release during gastrointestinal pH transition.

A concise HPLC-DAD method was developed and validated to support formulation characterization, ensuring selective and reliable quantification of both APIs. The method demonstrated ICH-compliant linearity, accuracy, and precision, and was applied to determine drug loading, extraction efficiency, and dissolution samples. Stability and solubility assessments confirmed suitability for formulation-specific quality control.

Overall, this work demonstrates the feasibility of producing lamivudine–zidovudine tablets via SSE 3D printing and highlights the potential of semi-solid extrusion and polymeric coatings for designing customizable antiretroviral dosage forms with tunable release characteristics.

Keywords: Semi-solid extrusion 3D printing, Lamivudine-zidovudine, Rheology, Controlled release, Eudragit coating, HPLC-DAD

1. Introduction

Oral drug administration remains the preferred route for many therapies due to its convenience, non-invasive nature, and generally favorable patient acceptability. This is important for chronic treatments, where long-term adherence and patient-centred dosing strategies are critical (Alqahtani et al., 2021; Stegemann et al., 2016). Conventional tablet manufacturing is based on well-established methods suitable for large-scale production. Despite their robustness and widespread industrial use, these processes offer limited flexibility when rapid changes in dose, dosage-form geometry, or drug-release characteristics are required. Three-dimensional printing provides an alternative manufacturing strategy that combines digital design with layer-by-layer fabrication, allowing greater control over the size, shape, internal structure, drug content, and release profile of the final dosage form (Norman et al., 2017) (Auel et al., 2025) . Moreover, the possibility of incorporating multiple active pharmaceutical ingredients within a single printed system creates opportunities for personalized fixed-dose combinations and polypill formulations (Khaled et al., 2015). Consequently, 3D printing represents a versatile platform for complementing conventional mass manufacturing with more flexible and patient-specific approaches to pharmaceutical production. Among 3D printing techniques, semi-solid extrusion has attracted increasing attention for the preparation of oral dosage forms, allowing the processing of drug-loaded semi-solid inks under mild conditions (Aina et al., 2025).

This feature is particularly advantageous for thermosensitive APIs and complex polymeric matrices, where conventional thermal processing may induce degradation or structural instability. Despite major advances in antiretroviral therapy, HIV infection remains a significant global health challenge, while combination antiretroviral therapy has transformed it from a rapidly progressive, life-threatening disease into a manageable chronic condition (Volberding & Deeks, 2010). Among the available therapeutic approaches, fixed-dose combinations have played an important role in simplifying treatment schemes, reducing pill burden, and improving patient adherence (Deeks et al., 2013). Lamivudine and zidovudine are nucleoside reverse transcriptase inhibitors that have been widely used as part of antiretroviral therapy. Their combination was introduced as Combivir®, one of the earliest fixed-dose combination products for HIV treatment, following the approval of lamivudine as a single agent. Combivir® contains

150 mg lamivudine and 300 mg zidovudine per film-coated tablet and are conventionally manufactured as oral solid dosage forms.

The selection of suitable excipients is therefore central to the successful development of semi-solid extrusion printed tablets. HPMCAS is a widely used pH-responsive enteric polymer in oral solid dosage forms, which enables modulation of drug release(Deshpande et al., 2018). To the best of our knowledge, the incorporation of HPMCAS-MF into SSE 3D-printed tablets has not previously been reported. Accordingly, the present study introduces a clear element of novelty by demonstrating, for the first time, the development and use of HPMCAS MF-based semi-solid inks for the fabrication of lamivudine–zidovudine combination tablets via SSE-3D printing.

In the present work, Carbopol 974P NF, gelatinized corn starch, and hydroxypropyl methylcellulose acetate succinate were combined to develop printable semi-solid inks. Carbopol was selected for its hydrogel-forming and mucoadhesive properties, while gelatinized starch contributed to the formation of a printable semi-solid matrix (Chatzitaki et al., 2023). HPMCAS was incorporated due to its pH-dependent solubility and potential contribution to modified release behavior, particularly under conditions simulating gastrointestinal transit. Based on formulation screening, the composition consisting of 10% Carbopol 974P NF, 70% corn starch, and 20% HPMCAS, at a starch ratio of 1:5, was selected as the most suitable ink for further development.

Although lamivudine and zidovudine are well-established antiretroviral agents and are commercially available as fixed-dose film-coated tablets, their combined incorporation into semi-solid extrusion 3D printed tablets has not been adequately explored. To date, no studies have reported the fabrication of lamivudine–zidovudine combination tablets using semi-solid extrusion, nor the development of an analytical platform specifically tailored to the characterization of such printed matrices. The development of such a platform could provide a flexible approach for producing combination antiretroviral dosage forms with tunable composition, geometry, and release characteristics. Furthermore, integrating formulation design with a validated analytical workflow is essential for ensuring reliable quality control of 3D-printed systems, which often present unique structural and compositional challenges compared with conventionally manufactured tablets. The use of polymeric matrices and post-printing coating strategies may offer additional control over drug release, especially during the transition from acidic gastric conditions to intestinal pH. A validated analytical method was also required to support the characterization of the printed matrices and ensure reliable quantification of both APIs.

Therefore, the aim of the present study was to develop 3d-printed lamivudine/zidovudine tablets using semi-solid extrusion 3D printing and to evaluate their physicochemical, mechanical, analytical, and biopharmaceutical performance. In addition to addressing the lack

of SSE-based antiretroviral combination tablets, the present work further introduces novelty by demonstrating, for the first time, the suitability HPMCAS for the development of semi-solid extrusion inks intended for 3D-printed oral dosage forms. For this purpose, printable blank and drug-loaded inks were prepared and characterized in terms of rheological behavior and printability. The printed tablets were lyophilized, subjected to physical quality control, and further evaluated by ATR-FTIR spectroscopy to assess drug incorporation and possible interactions with the excipients. In addition, an HPLC-PDA method was developed and validated for the simultaneous quantification of lamivudine and zidovudine and was subsequently applied to determine drug content and in vitro release performance. Finally, uncoated and Eudragit-coated tablets were compared in order to investigate the effect of coating level on the release kinetics and release mechanism of both active pharmaceutical ingredients.

2. Materials and methods

2.1. Materials

Carbopol 974P NF and Eudragit® L100-55 (Mw 135 kDa; apparent viscosity 50–200 mPa·s) were supplied by Lubrizol (Brussels, Belgium) and Evonik (Essen, Germany), respectively. Corn starch and hydroxypropyl methylcellulose acetate succinate (HPMCAS) were kindly donated by Pharmathen S.A. (Greece). Polyethylene glycol BioUltra (PEG 400) was purchased from Sigma-Aldrich (Steinheim, Germany). Lamivudine (99.93%) and zidovudine (98%) were obtained from AmBeed (Arlington Hts, IL, USA) and BLD Pharmatech (Reinbek, Germany), respectively. Milli-Q water and analytical-grade solvents were used throughout the study.

2.2. Preparation of printing inks

The developed semi-solid inks were formulated using Carbopol 974P NF, HPMCAS-MF, and gelatinized corn starch as the main polymeric/excipient components. Based on the preliminary formulation screening, the optimum composition consisted of 10% Carbopol 974P NF, 70% corn starch, and 20% HPMCAS-MF, expressed as % w/w relative to the total solid content. A starch:PBS ratio of 1:5 was used in order to provide the required moisture content and obtain a printable semi-solid formulation. Initially, HPMCAS-MF was dispersed in PBS under continuous stirring at ambient temperature for approximately 2 h, followed by corn starch gelatinization in a water bath at 86 ± 1 °, for 20 min under continuous stirring (Khaled et al., 2014). After gel formation, Carbopol 974P NF was incorporated into the mixture, followed by thorough stirring until complete homogenization was achieved. The resulting semi-solid ink was loaded into a dedicated printing syringe and stored overnight at 4 °C to allow adequate polymer hydration before the printing process.

Drug-loaded inks were prepared following the same procedure described for the blank ink, with the only modification being the encapsulation process. Specifically, lamivudine and zidovudine

were added to PBS at a 1:2 mass ratio, corresponding to that used in the commercially available formulation. The drug-containing PBS solution was then used for the addition of HPMCAS-MF, after which all subsequent preparation steps were carried out following the same procedure as for the blank ink, including filling of the printing syringe and overnight storage at 4 °C.

2.2. Development of the printing process of pharmaceutical dosage forms

The tablets were fabricated by semi-solid extrusion (SSE) three-dimensional printing using a XYZprinting 3D printer 3C10A (Taipei, Taiwan). The printing process was conducted at ambient temperature using a syringe nozzle of 2 mm diameter. The tablet design was created using AutoCAD 2019 (Autodesk Inc., CA, USA), while model slicing was performed using the Foodware software provided by the printer manufacturer. The final 3d-printed tablets were lyophilized and stored until further studies.

A portion of the produced tablets was further processed by applying a coating using the dip-coating method. Initially, a coating solution containing Eudragit L 100-55 and polyethylene glycol (PEG 400) was prepared. PEG 400, acting as a plasticizer, was selected instead of triethyl citrate (TEC), as it has been reported to be more effective by providing a lower glass transition temperature (Tg)(Khodaverdi et al., 2012). More specifically, 3 g of Eudragit L 100-55 were dissolved in 40 mL of an isopropanol–acetone mixture (1:1) under stirring at ambient temperature. Once a completely clear solution was obtained, PEG 400 was added at 10% w/w relative to the coating polymer and the solution was stirred for 60 min. Each tablet was immersed in the coating solution for 5 s, followed by a thermal curing step at a controlled temperature of 40 °C in an oven for 2 h, allowing sufficient time for the formation of the required coating film. Coatings consisting of three and six layers were prepared. After completion of each thermal curing step, the tablets were re-immersed in the coating solution to form the subsequent coating layer, and the process was repeated accordingly.

2.3. HPLC analysis method

For the pretreatment of the proposed pharmaceutical formulation, approximately 0.12 g of the powdered tablet was extracted with 20 mL of MeOH containing 0.2% FA (Formic Acid) and the mixture was centrifuged at 4,500 rpm for 15 minutes. The resulting supernatant was then diluted 1:200 with MeOH/H_2O (0.2% FA) and subjected to instrumental analysis. The determination of LAM and ZID was carried out on a Shimadzu HPLC system (Kyoto, Japan) equipped with a UV-DAD detector. Chromatographic separation was performed on an Aqua C18 Evosphere Fortis® column (4.6 × 250 mm, 5 μm). A binary mobile phase consisting of solvent A (H_2O with 0.1% formic acid) and solvent B (MeOH) was employed under a gradient elution program at a flow rate of 1 mL/min. The proportion of solvent B was increased from 10% to 30% at 4 min, returned to the initial composition at 10 min, and the system was allowed

to equilibrate for an additional 7 min. The column temperature was maintained at 45 °C, and the injection volume was set at 30 μL. Detection was performed at 280 nm for LAM and 265 nm for ZID. Additional analytical evaluations, including, solubility and stability assessments of both APIs, as well as excipient interference and system-suitability testing, were also performed to support the reliability of the chromatographic method.

2.4. Physicochemical characterization of the inks and three-dimensional printed tablets

2.4.1. Evaluation of the rheological properties of blank and drug-loaded inks

The rheological properties of the inks were investigated using an MCR 302 rotational rheometer (Anton Paar, Graz, Austria) after storage at 4 °C overnight. Measurements were performed at ambient temperature using a parallel-plate geometry with a plate diameter of 25 mm and a fixed gap of 1 mm. Initially, the viscosity of the inks was determined as a function of shear rate over a range of 0.1–1000 s^{-1}. The inks were also subjected to an initial strain amplitude sweep over a strain range of 0.01–100% in order to identify the linear viscoelastic region and define the shear strain value, γ, to be used in the subsequent measurements. To assess the viscoelastic behavior of the inks, frequency sweep tests were performed over an angular frequency range of 1–1000 rad/s at a constant shear strain of $\gamma = 0.1\%$. This strain value was selected to ensure that the measurements were conducted within the linear viscoelastic region, thereby preserving the structural integrity of the samples during testing. Finally, the ability of the inks to recover their elastic behavior after exposure to high shear was evaluated ωια a three-interval thixotropy test (3ITT), during which the viscosity of the inks was recorded as a function of time. The first and third intervals were conducted under a low shear rate of $\dot{\gamma} = 1\ s^{-1}$, whereas the second interval was performed under a high shear rate of $\dot{\gamma} = 100\ s^{-1}$. Recovery of the initial viscosity after printing is a critical factor for maintaining the morphological integrity and shape fidelity of the printed tablets.

2.4.2. Attenuated total reflectance Fourier-transform infrared spectroscopy (ATR-FTIR) of 3d-printed tablets

ATR-FTIR spectra were acquired using a Shimadzu IR-Prestige-21 FT-IR spectrophotometer (Shimadzu Europa GmbH, Duisburg, Germany) equipped with a high-throughput Golden Gate MKII single-reflection ATR accessory (Specac, Kent, UK) fitted with ZnSe lenses. Spectra of the raw materials and lyophilized tablets were recorded to investigate potential intramolecular interactions between the active pharmaceutical ingredients and the polymeric carrier. Samples were analyzed over a wavenumber of 4000-600 cm^{-1} using 64 scans at 4 cm^{-1} resolution. Spectra were acquired and processed using IRsolution software (Shimadzu, Kyoto, Japan).

2.5. Quality assessment of 3d-printed tablets

The 3d-printed tablets were evaluated in terms of weight, dimensions, friability, and disintegration.

2.5.1. Evaluation of tablet weight and dimensions

The weight of 10 lyophilized tablets from each batch, including blank tablets, drug-loaded uncoated tablets, and tablets coated with three or six coating layers, was measured. The mean weight and standard deviation were calculated for each tablet type. In addition, tablet thickness and diameter were measured using a digital Vernier caliper with an accuracy of 0.01 mm.

2.5.2. Friability evaluation of 3d-printed tablets

Tablets with a total weight of 6.5 g were placed in an Erweka friabilator and rotated at 25 rpm for a total duration of 4 min. The tablet weight was recorded after completion of the test, and the percentage weight loss was calculated.

2.5.3. Tablet disintegration test

The disintegration test was performed in accordance with pharmacopoeia requirements (USP/Ph. Eur.). The tablets were initially immersed in an acidic medium consisting of 0.1 M HCl for 60 min at 37 ± 2 °C. Subsequently, the tablets were quantitatively transferred to a phosphate buffer medium at pH 6.8, where they remained for 2 h at the same temperature.

2.6. Quantitative determination of the entrapped LAM and ZID

To determine the drug loading of lamivudine and zidovudine in the lyophilized tablets, the previously described recovery–extraction procedure was applied using MeOH containing 0.2% FA as the extraction solvent. The procedure was performed on accurately weighed amounts of powdered tablet samples, and the drug loading percentages of lamivudine and zidovudine (%DL) were calculated according to Equation 1.

$$\text{Encapsulation Efficiency (\%)} = \frac{\text{Amount of encapsulated API (mg)}}{\text{Theoretical or total amount of API (mg)}} \times 100 \qquad \{1\}$$

2.7. *In vitro* release study of the active pharmaceutical ingredients from the three-dimensional printed tablets

The *in vitro* release of LAM and ZID was investigated using a USP II dissolution apparatus at a constant temperature of 37 ± 0.1 °C and a paddle rotation speed of 50 rpm. The tablets, including uncoated tablets and tablets coated with three or six coating layers, were immersed in 900 mL of 0.1 N HCl for 2 h to evaluate their resistance under acidic conditions. Subsequently, they were quantitatively transferred to pre-warmed simulated intestinal fluid (SIF, pH 6.8), and the release study was continued for an additional 6 h.

At predetermined time intervals up to 8 h, 5 mL samples were withdrawn and immediately replaced with an equal volume of pre-warmed dissolution medium. The collected samples were centrifuged at 4500 rpm for 15 min, and an aliquot of the supernatant was transferred to HPLC vials after dilution with MeOH at a 1:1 v/v ratio.

2.8. Release kinetics and release mechanism of LAM and ZID from the 3d-printed tablets

The release profiles of LAM and ZID from the printed tablets were analyzed using different mathematical models in order to determine the release kinetics and the underlying release mechanism. Specifically, zero-order, first-order, Korsmeyer–Peppas, Higuchi, Hixson–Crowell, Hopfenberg, and Weibull models were evaluated. The most suitable and representative model was selected based on the highest coefficient of determination (R^2), indicating the best fit to the experimental data. Data processing and analysis were performed using DDSolver, an Excel add-in that enables the application and comparison of multiple dissolution kinetic models.

3. Results and discussion

3.1. Chromatographic Strategy and Method Development

Several HPLC-UV/DAD methods have been reported for the determination of LAM and ZID, many of which employ short chromatographic runs ranging from 0.5 to 7 minutes (Erkmen et al., 2019; Panda et al., 2022; Vieira-Sellaï et al., 2022). However, the development of an ultrafast method was not appropriate for the present study. The pharmaceutical product was manufactured through a three-dimensional (3D) printing process, a technique that may induce thermal or process-related degradation and generate minor byproducts or structural modifications, while the excipient composition of the printed formulation could introduce additional chromatographic interferences. For this reason, a method with sufficient chromatographic space was required to allow the detection of potential degradation products or excipient-derived impurities that would likely co-elute under very short gradients.

To achieve this, the method was designed to begin with a low percentage of organic solvent, exploiting the retention characteristics of the Aqua C18 column, thereby enabling effective separation of the hydrophilic excipients from LAM. Subsequently, a gradient increase in organic content was applied to maintain a relatively short overall analysis time while ensuring adequate resolution of more lipophilic excipients from ZID.

3.2. Method validation

A series of preliminary experiments (Table S1) were conducted for the development of the proposed method. The optimized method was validated in accordance with ICH guidelines in terms of selectivity, linearity, precision and accuracy. The performance, applicability and simplicity of the analytical procedure was also evaluated with specific analytical assessment

indexes. Selectivity was investigated by analyzing a blank diluent and a blank unspiked tablet pretreated sample. No interfering signals were observed at the retention time of the APIs, demonstrating the high selectivity of the method (Fig. S1). Potential carryover effects were examined by injecting blank samples immediately after the analysis of three samples at the highest calibration level (25 µg/mL). Linearity was evaluated through calibration curves (r2=0.999) generated for the two analytes across six concentration levels (0.25 to 25 µg/mL). Calibration curves and the analytical suitability statistics (n=3) are summarized in Table. S2. The limits of detection (LODLAM: 0.13 µg/mL; LODZID: 0.11 µg/mL) and quantification (LOQLAM: 0.39 µg/mL; LOQZID: 0.33 µg/mL) were calculated as 3.3 and 10 times the ratio of slope/standard deviation (SD), respectively.

Method accuracy and precision were determined through intra-day (n = 3) and inter-day (three consecutive days) studies conducted at three concentration levels. The %RSD values were consistently below 2.5 % for both intra-day and inter-day measurements. Additionally, % recovery values feel within the range of 95.9 to 102.1%, verifying the method's precision, and accuracy (Table S3). Finally, the robustness of the HPLC conditions was evaluated through peak area and shape (Tailing Factor), regarding the concentration of FA, the initial content of MeOH, the flow rate of the mobile phase, the oven temperature and the wavelength. The method was evaluated as robust (%RSD < 2).

The method is characterized by its performance, efficiency, practicality, simplicity and applicability (Figure S2) (Manousi et al., 2023; Mansour et al., 2025; Nowak et al., 2025).

3.3. Sample pretreatment

A Solid-Liquid Extraction (SLE) procedure was selected for tablet pretreatment. The choice of extraction solvent was based on the requirement that both active pharmaceutical ingredients (LAM and ZID) are readily soluble in methanol and acetonitrile, whereas the formulation excipients are not. To verify this selectivity, a preliminary assessment was performed using individual excipients as well as the un-spiked pharmaceutical formulation, following the proposed pretreatment protocol (Table S4) (Godela & G, 2020; Yang et al., 2026). The optimized cleanup method was evaluated in five independent replicates, yielding mean recoveries (%RSD) of 96.4% (0.89%) for LAM and 98.3% (0.97%) for ZID, thereby confirming the suitability and robustness of the extraction procedure.

The stability and solubility behavior of lamivudine and zidovudine under conditions relevant to tablet preparation, extraction, and chromatographic analysis was systematically evaluated to ensure the reliability of the analytical workflow and the integrity of the APIs throughout processing.

Short-term stability was assessed in the selected extraction solvent (MeOH containing 0.2% FA), demonstrating that both APIs remained chemically stable over a 24-hour period, with peak

area variation below 1.2% for LAM and 1.0% for ZID (n = 3). No additional chromatographic peaks or spectral deviations were detected, confirming the absence of degradation phenomena. Solubility studies verified complete dissolution of both APIs in the extraction medium, with measured solubility values exceeding 50 mg/mL for lamivudine and 20 mg/mL for zidovudine in MeOH/H_2O mixtures. These values ensured quantitative extraction from the printed matrices and excluded the possibility of solubility-related losses during sample handling.

3.4. Rheological analysis of the inks

The determination of ink viscosity is of major importance for achieving satisfactory printability, while rheological analysis represents a critical preliminary assessment for selecting the most suitable ink formulation for the subsequent studies. As shown in Figure 1a, all inks exhibited shear-thinning behavior, as their viscosity progressively decreased with increasing shear rate. This behavior is particularly desirable in semi-solid extrusion three-dimensional printing, as it facilitates material flow during extrusion (del-Mazo-Barbara & Ginebra, 2021).

The incorporation of LAM and ZID into the ink did not disrupt the polymeric network of the system, as the drug-loaded ink displayed a similar rheological profile, with only slightly increased viscosity values. This finding further supports the suitability of the selected formulation in terms of both rheological behavior and printability.

The amplitude sweep test was used to determine the viscoelastic properties of the inks and a shear strain of 0.1% was selected for the frequency sweep measurements, as this value lies within the linear viscoelastic region. The mechanical strength of the inks was then evaluated by oscillatory frequency sweep measurements, as presented in Figure 1b. A characteristic feature of printable gel-like inks is that the storage modulus, G′, is higher than the loss modulus, G″, which was clearly confirmed by the obtained results. G′ represents the elastic component of the viscoelastic response and is associated with solid-like behavior, whereas G″ corresponds to the viscous component and reflects liquid-like behavior (Chatzitaki et al., 2022).

The ability of the inks to recover their initial viscosity after the application of high shear is also a critical parameter for assessing ink suitability and the capacity of the final printed tablets to retain their structural integrity. As shown in Figure 1c, the application of high shear, simulating extrusion through the printing nozzle, caused a pronounced viscosity decrease, indicating disruption of the polymeric network and facilitating smooth flow. Once the shear was reduced, viscosity rapidly recovered to near-initial values, supporting shape retention and structural stability after deposition. The incorporation of LAM and ZID did not substantially alter the rheological behavior compared with the corresponding blank formulations.

3.5. Characterization of the 3d-printed tablets by Fourier-transform infrared spectroscopy (FT-IR)

The FT-IR spectral analysis (Figure 1d) provides important structural information, from the individual raw materials to the final formulated tablets. In all ink formulations, the broad band observed at approximately 3300 cm^{-1} was attributed to the formation of intermolecular hydrogen bonds between the polymers and water, supporting hydrogel formation. The reduced intensity of this band in the lyophilized tablets can be attributed to dehydration resulting from the freeze-drying process. In both blank tablets and drug-loaded tablets, either uncoated or coated, the spectra were dominated by the absorption bands of the polymeric excipients, including the carbonyl band at approximately 1650 cm^{-1} and the C–O stretching band at approximately 1100 cm^{-1}. The characteristic bands of APIs were not detected, which may be associated with drug amorphization, their dispersion within the polymeric matrix, or the sensitivity limits of the analytical technique. This behavior indicates the successful incorporation and distribution of the APIs within the tablet matrix and suggests good chemical compatibility among the formulation components. Finally, tablets coated with six layers exhibited a noticeably higher intensity of the band in the 1650–1700 cm^{-1} region, corresponding to C=O stretching, compared with tablets coated with three layers. This confirms the higher extent of Eudragit deposition achieved through the additional coating layers (Li et al., 2003; Marcos et al., 2018; Ramkumaar et al., 2012; Solís-Cruz et al., 2023)

3.5. Physical quality control and characterization of the three-dimensional printed tablets

The effective incorporation of LAM and ZID into the 3d-printed tablets was confirmed, with entrapment values of 96.19 ± 2.90% and 103.11 ± 1.61%, respectively. These results indicate that the overall process, from ink development and printing to final tablet recovery, did not adversely affect the APIs. The 3d-printed tablets were characterized in accordance with USP requirements in terms of weight, dimensions, friability, and disintegration.

Based on the tablet weight and the achieved entrapment levels, each tablet contained 28.18 ± 0.85 mg of LAM and 60.42 ± 0.94 mg of ZID. The coating process did not induce a statistically significant change in the entrapment percentage of the active substances, suggesting that this post-processing step did not alter the drug content within the tablet matrix.

The friability test showed no weight loss, thereby satisfying the relevant USP requirement. During the disintegration test, the examined tablets remained intact in the acidic medium, whereas in phosphate buffer, the tablets swelled but did not fully disintegrate within the predefined 2 h period, while the core remained hard.

3.7. *In vitro* release study of lamivudine and zidovudine from three-dimensional printed tablets

The results of the release study for the LAM and ZID, under conditions simulating the transit of the tablets through the gastrointestinal tract, were illustrated in Figure 2. The uncoated tablets exhibited a rapid release profile, characterized by an initial burst effect. LAM, owing to its higher hydrophilicity, exhibited a rapid release profile, reaching approximately 80% release within the first 2 h in 0.1 N HCl, followed by complete release after 240 min in SIF at pH 6.8. Despite the gastro-resistant/enteric properties of HPMCAS and the ability of Carbopol to absorb water within its polymeric chains, leading to tablet swelling, no substantial restriction of drug diffusion was observed during the early time points (Tran et al., 2021). This behavior could be attributed to the porous structure formed after lyophilization, which may facilitate the penetration of the dissolution medium into the tablet matrix and consequently enhance the diffusion of the APIs (Ferrero et al., 2010; Jange et al., 2023). The coating offered protection in the acidic medium, resulting in a slower and more controlled release of both APIs. The release rate was affected by the number of coating layers, as the increase from three to six layers further reduced drug release, particularly during the first 2 h in acidic medium. The lowest release percentage was observed for ZID-loaded tablets coated with six layers, reaching 20.50% after 2 h in acidic medium. Although the coating did not completely prevent drug release, it formed a barrier that limited the diffusion of the APIs through the tablet matrix. The coating level was approximately 16.48% and 33.01% for tablets coated with three and six layers, respectively, as calculated from the tablet weight increase after coating. At the end of the 8 h dissolution experiment, all formulations showed more than 90% release of both APIs. Overall, LAM exhibited a faster release rate compared to ZID, which could be attributed to differences in their physicochemical properties, particularly aqueous solubility and diffusion behavior.

The release profiles of both APIs followed similar kinetic trends, as shown in Table 2. Based on the coefficients of determination (R^2), the Korsmeyer–Peppas model provided the best fit for the uncoated tablets for both APIs. In contrast, for the coated tablets, regardless of the coating level, the Weibull model showed the strongest correlation with the experimental data. This difference in model fitting suggests that the empirical polymeric matrix diffusion model better describes the release mechanism of the uncoated tablets, whereas the Weibull model more accurately captures the more complex and time-dependent release profile of the two active substances from the coated formulations, particularly for $\beta > 1$ (Papadopoulou et al., 2006).

4. Conclusions

In the present study, LAM and ZID-loaded tablets were successfully prepared using SSE 3D printing, while an HPLC-DAD method was developed and validated for their qualitative and quantitative determination. The method showed satisfactory chromatographic performance, complete separation of both APIs, and no interference from the excipients. The analytical method was successfully validated in terms of linearity, selectivity, accuracy, repeatability, and

robustness. Furthermore, UV, stability, and solubility studies were performed to support the analytical results.

The drug content of the 3D-printed tablets was 96.19% for LAM and 103.11% for ZID, confirming their successful incorporation into the polymeric matrix. ATR-FTIR analysis did not reveal any new characteristic bands, suggesting the absence of chemical interactions between the APIs and the excipients. The *in vitro* release study showed that the coating affected the release profile of both APIs. The uncoated tablets exhibited a faster release rate, with approximately 80% of LAM and 60% of ZID released within the first 2 h in acidic medium. Although Carbopol and HPMCAS have the ability to absorb water and swell, their combination was not sufficient to limit drug release in acidic conditions. This behavior could be attributed to the porous structure formed after lyophilization, which may facilitate the penetration of the dissolution medium into the polymeric matrix and consequently enhance drug diffusion.

The coated tablets showed a slower release rate compared to the uncoated formulations. Increasing the number of coating layers further reduced the release of both APIs, particularly during the acidic stage. These results indicated that the coating formed a barrier that limited drug diffusion through the tablet matrix. Furthermore, the release data were fitted to different kinetic models, including Korsmeyer–Peppas and Weibull, providing additional information about the release behavior of the developed tablets.

Overall, this study demonstrated the feasibility of producing LAM- and ZID-loaded tablets using SSE 3D printing. To the best of our knowledge, HPMCAS was used for the first time in SSE 3D printing, indicating its potential for future pharmaceutical applications. The developed HPLC-PDA method enabled the accurate determination of both APIs and the evaluation of their *in vitro* release profiles. In conclusion, the combination of suitable polymeric formulations, SSE 3D printing, and polymeric coating represents a promising approach for the development of pharmaceutical tablets with adjustable drug release characteristics.

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Petros Livanas: Methodology, Investigation, Formal analysis, Data curation. Aikaterini-Theodora Chatzitaki: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. Georgios Kamaris: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. Catherine Markopoulou: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Table 1. Physicochemical properties of the 3d-printed tablets.

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Table 2. Kinetic parameters of LAM and ZID release from uncoated and coated tablets.

		Zero order	First order	Higuchi	Korsmeyer-Peppas	Hixson-Crowell	Hopfenberg	Weibull
Uncoated tablets	LAM	R^2=2.095 K0=0.339	R^2=0.964 K1=0.015	R^2=0.521 KH=6.142	R^2=1.000 KKP=5.685, n=0.562	R^2=0.913 KHC=0.003	R^2=0.968 KHB=0.002, n=8.071	R^2=0.975 α=44081.518, β=2.113
	ZID	R^2=0.630 K0=0.306	R^2=0.9854 K1=0.0095	R^2=0.859 KH=5.464	R^2=0.999 KKP=5.316, n=0.526	R^2=0.956 KHC=0.002	R^2=0.9854 KHB=5.5221E-5, n=1711.8	R^2=0.995 α=1757.6, β=1.456
3 layers coating	LAM	R^2=0.262 K0=0.297	R^2=0.970 K1=0.008	R^2=0.907 KH=5.229	R^2=0.989 KKP=1.102, n=0.820	R^2=0.991 KHC=0.002	R^2=0.992 KHB=0.002, n=2.465	R^2=0.995 α=15308.173, β=1.845
	ZID	R^2=0.720 K0=0.261	R^2=0.938 K1=0.005	R^2=0.871 KH=4.492	R^2=0.977 KKP=0.069, n=1.325	R^2=0.962 KHC=0.001	R^2=0.963 KHB=0.002, n=2.196	R^2=0.902 α=1409.4, β=1.397
6 layers coating	LAM	R^2=0.696 K0=0.273	R^2=0.921 K1=0.006	R^2=0.855 KH=4.736	R^2=0.965 KKP=0.049, n=1.413	R^2=0.949 KHC=0.002	R^2=0.957 KHB=0.003, n=1.618	R^2=0.977 α=3995.912, β=1.617
	ZID	R^2=0.738 K0=0.258	R^2=0.864 K1=0.005	R^2=0.781 KH=4.408	R^2=0.955 KKP=0.001, n=2.120	R^2=0.894 KHC=0.001	R^2=0.906 KHB=0.003, n=1.360	R^2=0.959 α=430.374, β=1.242

Figures and legends

Figure 1. Rheological analysis of loaded and unloaded inks. (a) Viscosity measurements at shear rates 0.1 –1000 1/s, (b) storage (G') and loss (G'') modulus of frequency sweep test at strain value of 0.1% and frequency range of 1 to 1000 Hz, (c) ATR -FTIR spectra of raw materials and loaded and unloaded tablets.

Figure 2. *In vitro* release profiles of LAM and ZID from uncoated and tablets with different coating levels in pH 7.4.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ethics declaration

The author(s) declare(s) that the study does not involve humans nor animal subjects.